

of applications require databases to be faster and have a more predictable response. This has also led to databases that can process different kinds of inputs. There is a requirement for real-time analytics for stock market information—a ‘streaming’ input that is quite different from the traditional DB input. A good columnar in-memory DB will be required to handle this.

3. The usage of databases in different Web applications has meant that the nature of the data stored is very different. This applies to changes in the nature of data types, the relation between data and the nature of queries. A typical example could be *LinkedIn*, which stores profiles of people and links the many fields of one person to the many other fields of another person. If a traditional RDBMS schema is applied, it would be very difficult to manage it. A good document database will help this cause.

There are many more reasons for this evolution. For example, people no longer want to use SQL to operate databases. They would rather use the Web paradigm to operate the databases; so there is a need for databases to support PHP drivers and natively support Representational State Transfer (REST).

The increasing importance of open source in the DB space

Along with the transformation in the DB field, another critical aspect is that open source databases have led this transformation. Hadoop was one of the first databases to signal the transformation. Hadoop-Hbase has been rapidly followed by great open source new-generation databases. Today, in the emerging NoSQL field, open source databases rule the roost. This has also led the industry to take a more serious look at older open source databases, the most important being PostgreSQL. This is the ‘grand old man’ of all open source databases and in the past few years it has become more and more prominent. In this article, we will take a brief look at the open source offerings in various functional spaces of the DB panorama.

The ‘relational’ DB space

Relational databases (RDBMS) have been the de-facto DB standard for years. The DBMS emerged in the 1970s and has been the subject of a lot of academic research. Its foundation is strongly based on relational-algebra and offers a mathematical basis for its principles. This has made these databases very stable and often quite predictable. Relational databases have been put to use in two broad functional areas: Transaction Processing (OLTP) and Analytical Processing (OLAP).

OLTP: This has been the primary strength and the use of databases. OLTP databases offer transaction processing capability. An OLTP system comes into play when we access our bank account or book a train ticket. These databases require a very fast response and, at the same time, very

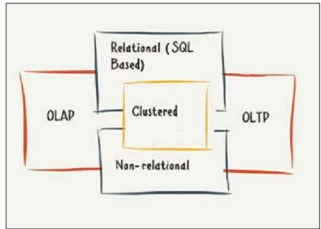


Figure 1: DB Space

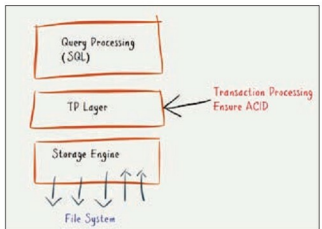


Figure 2: OLTP architecture

reliable operation. OLTP systems normally offer ACID (Atomicity, Consistency, Isolation and Durability). OLTP DBs are normally medium-sized (up to a few TB) and are generally disk-oriented. These systems generally handle a mix of reads and writes. A very brief conceptual diagram of an OLTP database is provided in Figure 2.

Postgres and MySQL are the dominant open source players in the OLTP-relational database space. These databases are very mature, like Postgres, which has been evolving over the years, and now offers better performance and features. It is easily one of the most functionally rich DBs in the world and can compare favourably with any of its commercial competitors. Many commercial and cutting-edge databases have been spun-off from Postgres -- a good example would be GreenPlum. MySQL (acquired by Oracle now) is another prominent open source database and has been extensively used for Web based applications. One of the most famous deployments of MySQL is Facebook.

Emerging OLTP: The OLTP space has remained unchanged for a long time. However, one of the recent strong trends has been to move the OLTP DB to memory for very high performance. This technology is one of the most advanced in